



University of Sheffield

The application of regularisation to common datasets in machine learning

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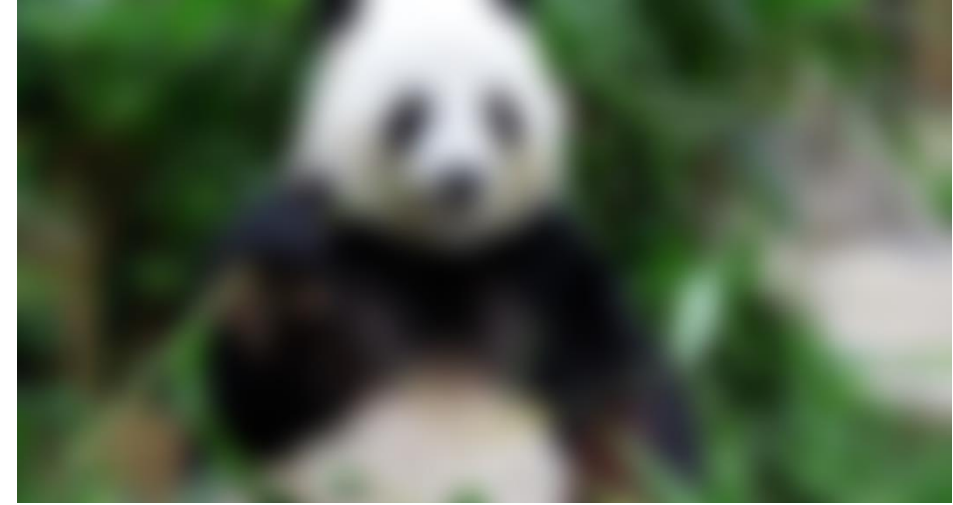
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The Problem

- Linear inverse problems involve moving backward from observed data (**b**) to determine the model parameters (**x**)

$$\mathbf{Ax} = \mathbf{b}$$

- What are some examples?
- Inverse problems are prone to ill-posedness
- What does ill-posed mean?
- Why is this problematic?



The Solution

- Regularisation is a process used to treat these ill-posed inverse problems
- How does this work?
- This underpins the **bias-variance trade-off**

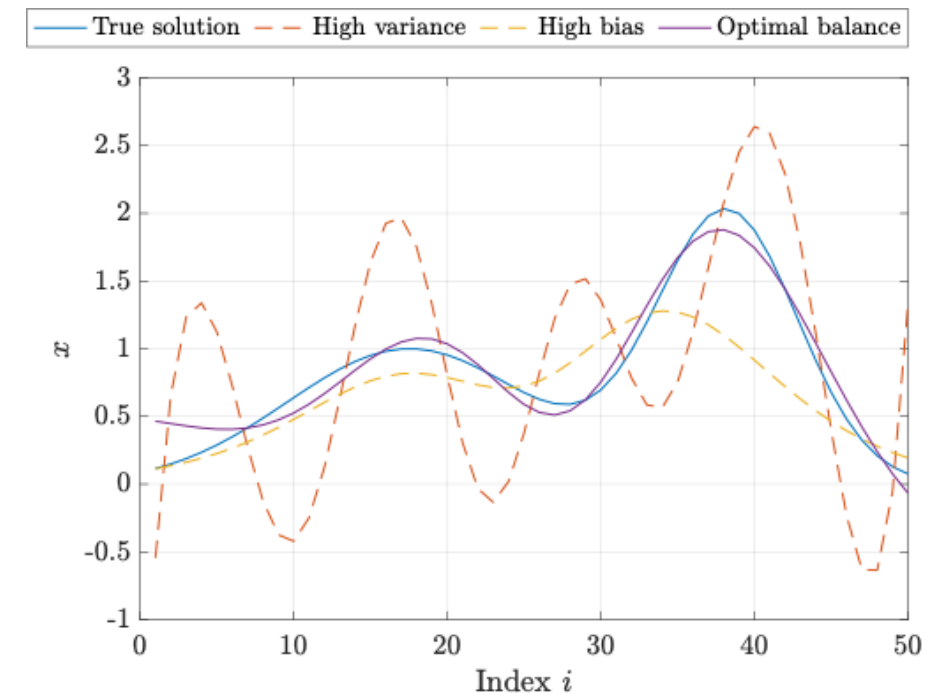
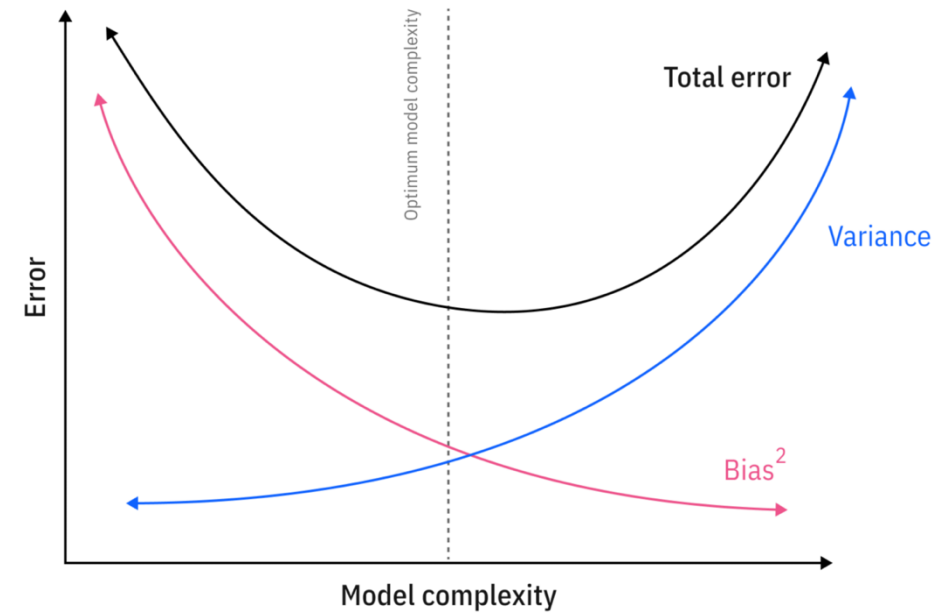
$$\text{MSE} = \text{bias}^2 + \text{variance}$$

- **The goal:**

Unusable system
dominated by noise



Stable, robust,
predictive model



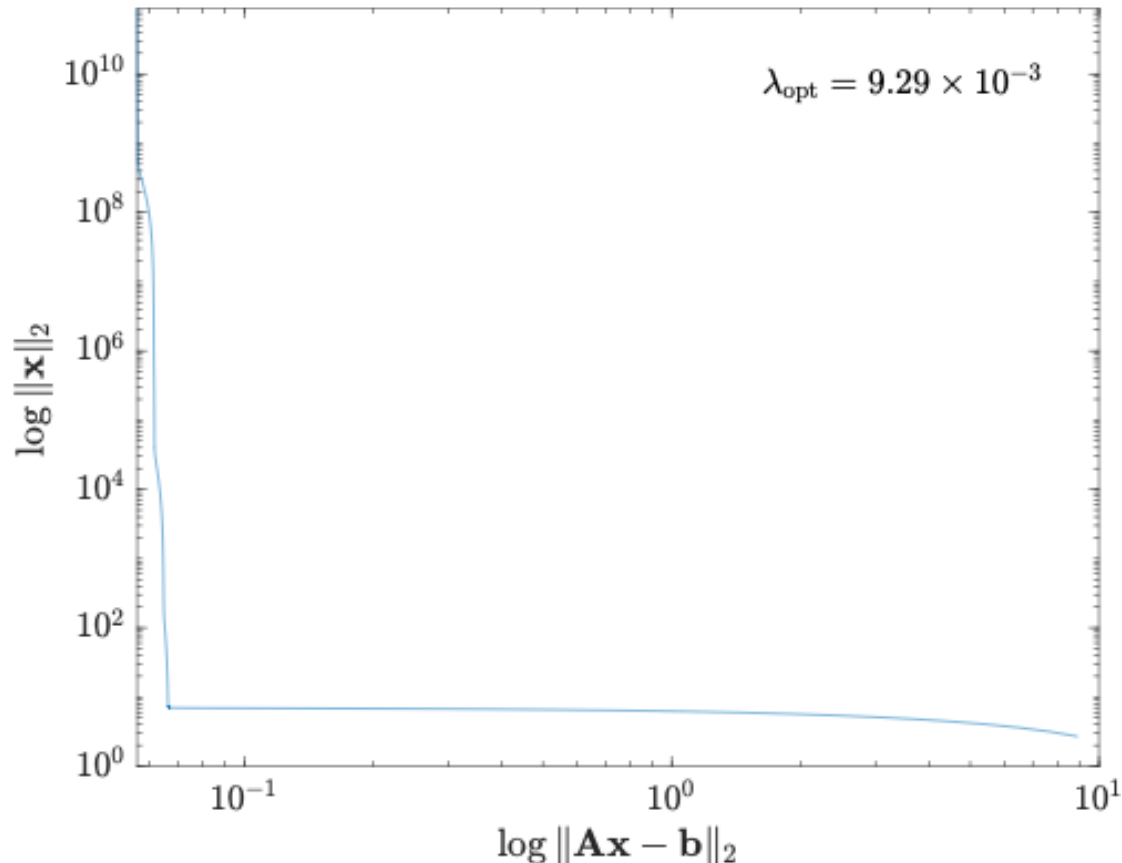
Motivation and Aims

- Regularisation is often treated like a “black-box” default that is blindly applied in machine learning pipelines without understanding why
- **Why does this matter?**
 - Application is not benign
 - Unnecessarily introduces error to an already stable system
 - Can suppress real features
- **Project aims:**
 - Define the conditions for which it is necessary
 - Test these conditions on common machine learning real-world datasets
 - Verify the regularised solution satisfies conditions on error and stability

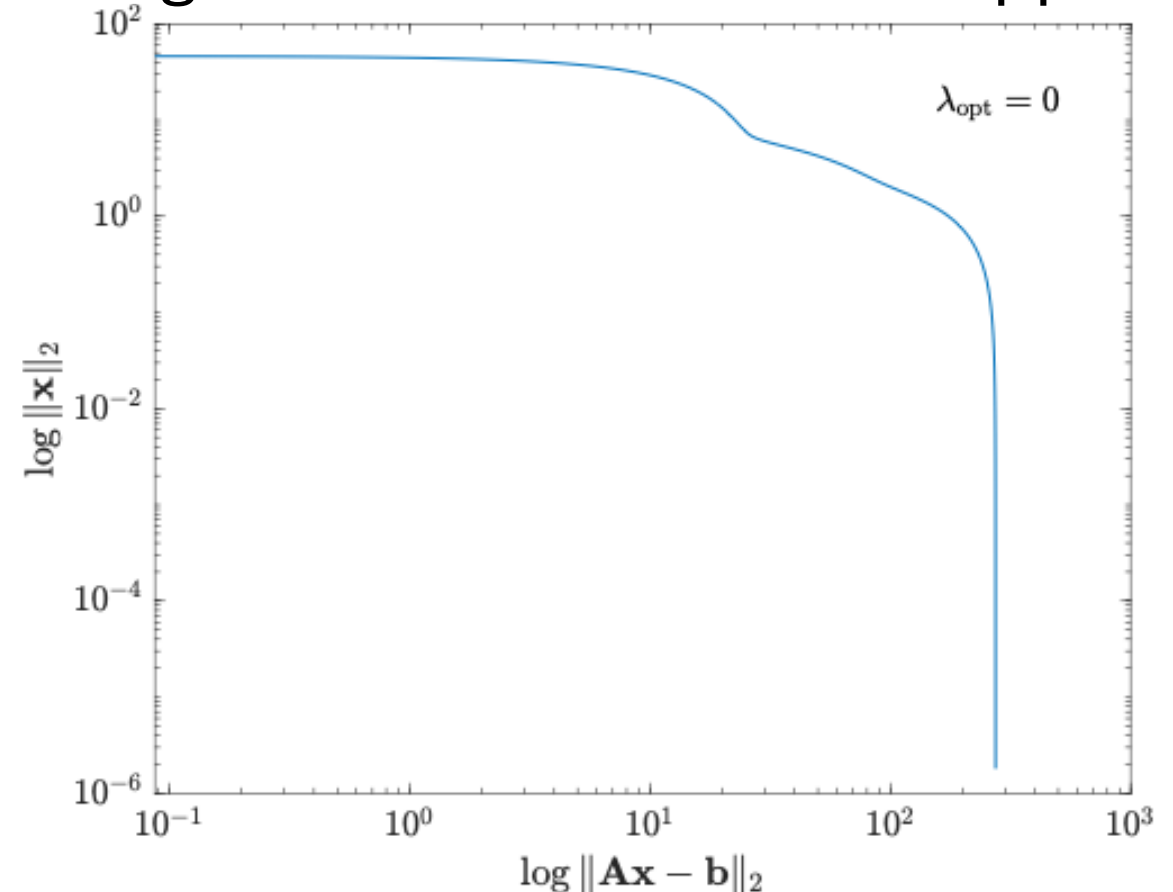
Mathematical Conditions for Regularisation

- The L-curve, solution norm against residual norm, is used to qualitatively diagnose whether regularisation is required

Apply regularisation



Regularisation must not be applied



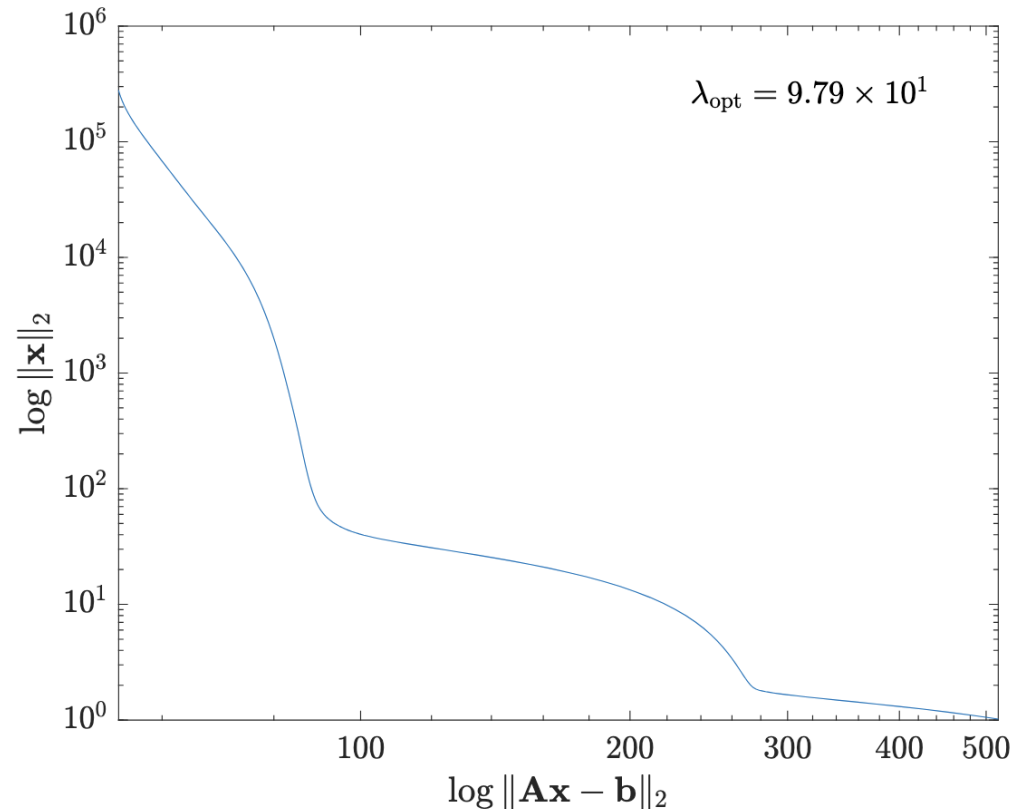
Materials and Methods

- **Framework:** Matlab R2022b using “regtools” package (Hansen 2007)
- **Experimental controls:**
 - 80/20 training/testing data split
 - Averaging over many trials with fixed seed generation for randomised trials
 - Features normalised for baseline comparison
- **Ill-posed – Tecator dataset (StatLib Datasets Archive 1995):**
 - 240 meat samples; 100 NIR spectral features predicting moisture content
 - Extreme multicollinearity and ill-conditioned
- **Well-posed – California housing dataset (Pace & Barry 1997):**
 - 20,640 samples; 8 demographic features predicting median house value
 - Stable, independent variables

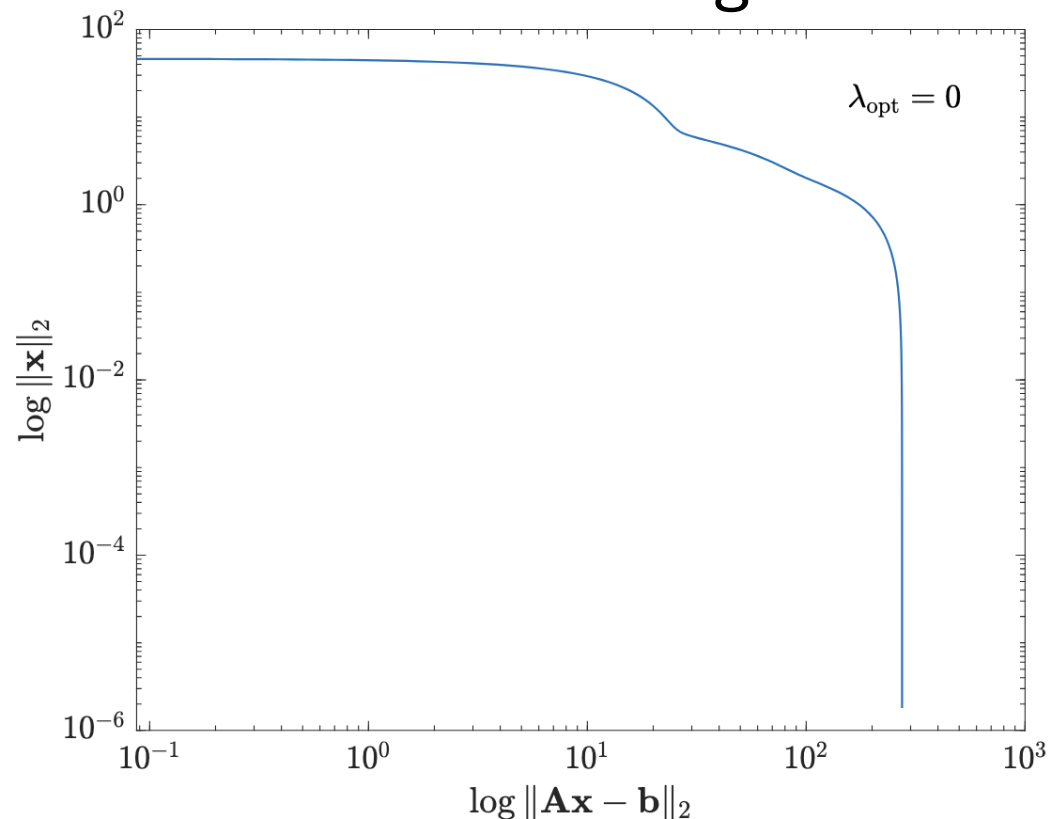
Real-World Dataset Diagnosis

- First step must be to diagnose the problem
- Once established, the ‘best’ regularisation technique was tested
- How was the ‘best’ technique determined?

Tecator dataset



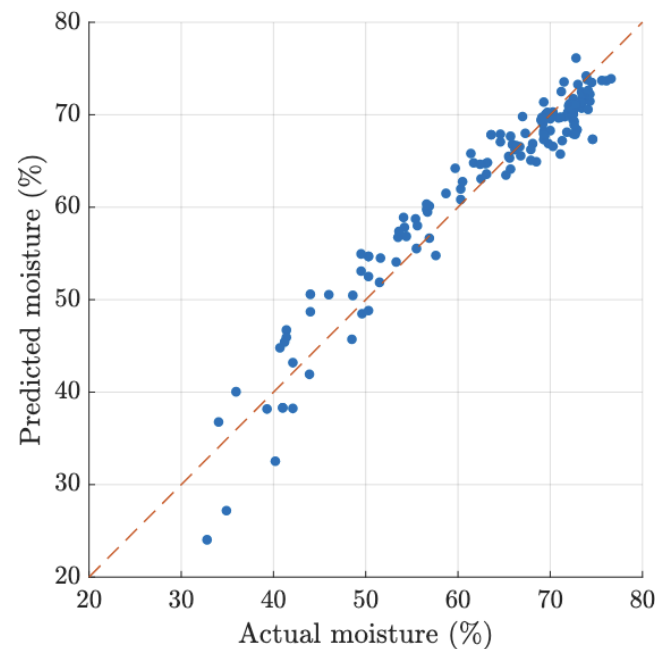
California housing dataset



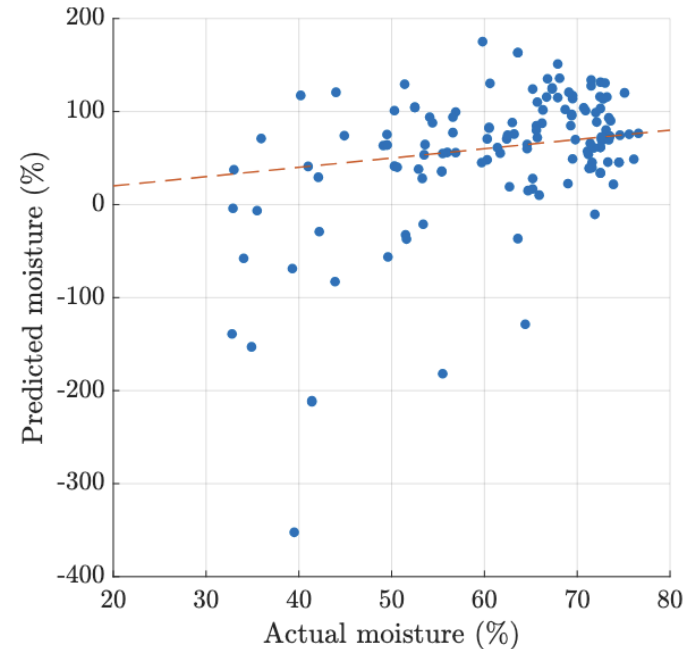
Comparison of Techniques

- Regularised solutions severely reduced error with a much higher stability
- The lasso was most accurate and stable whilst using the least features

Technique	Mean RMSE	Std. dev.	Mean features	Run time (s)
Unregularised	99.6778	74.1485	95.1	0.21
Tikhonov	6.1632	0.3920	100	6.09
TSVD	6.4127	0.7051	100	19.50
Elastic net	3.0836	0.2623	66	2600.51
Lasso	3.0765	0.2143	32.6	325.83



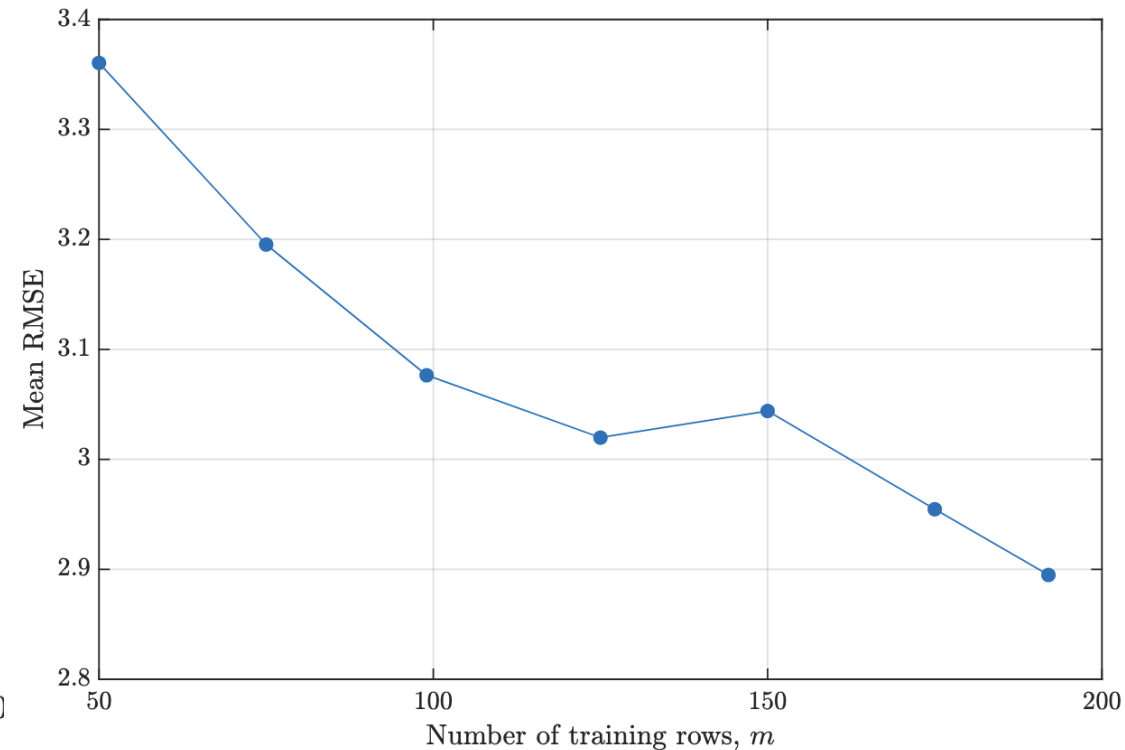
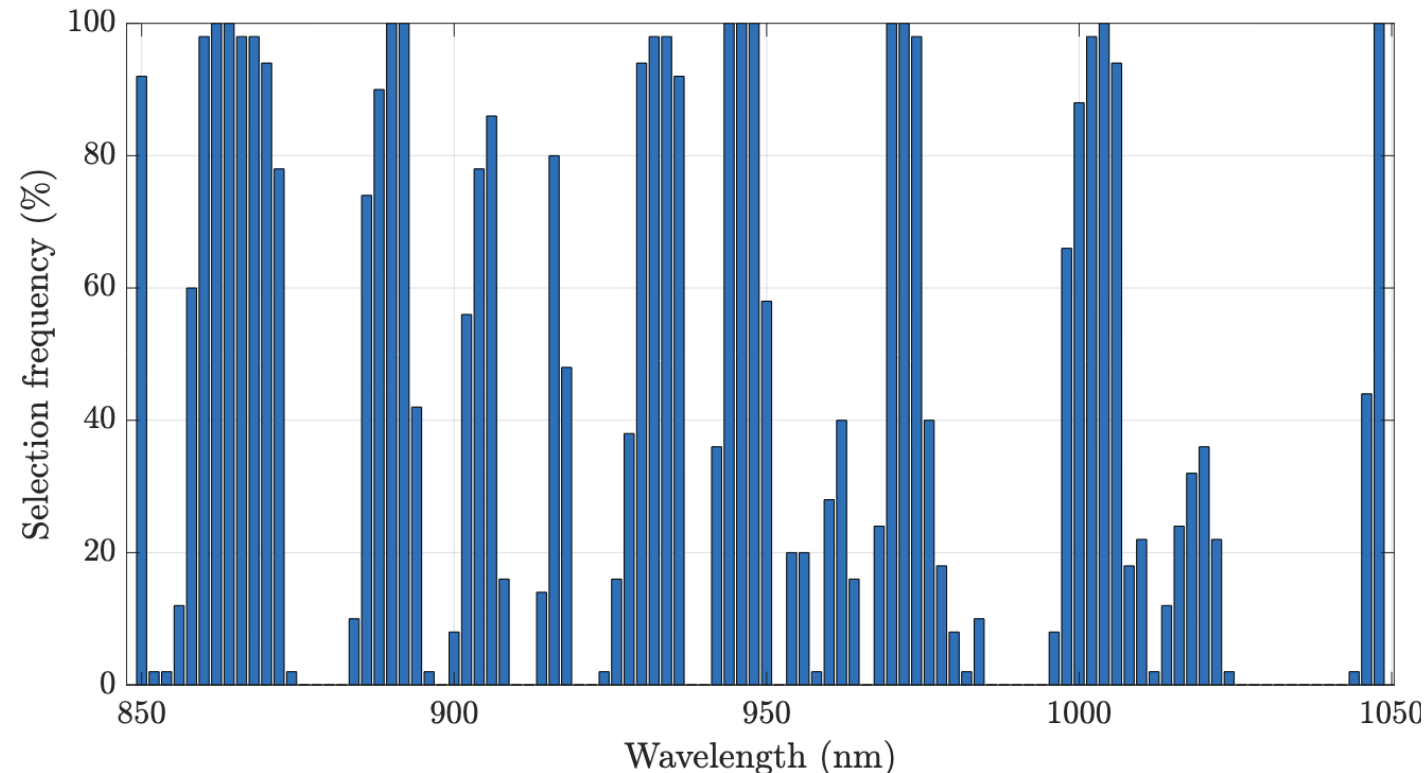
(a) Technique — Lasso.



(b) Technique — Unregularised.

Further Testing of the Lasso

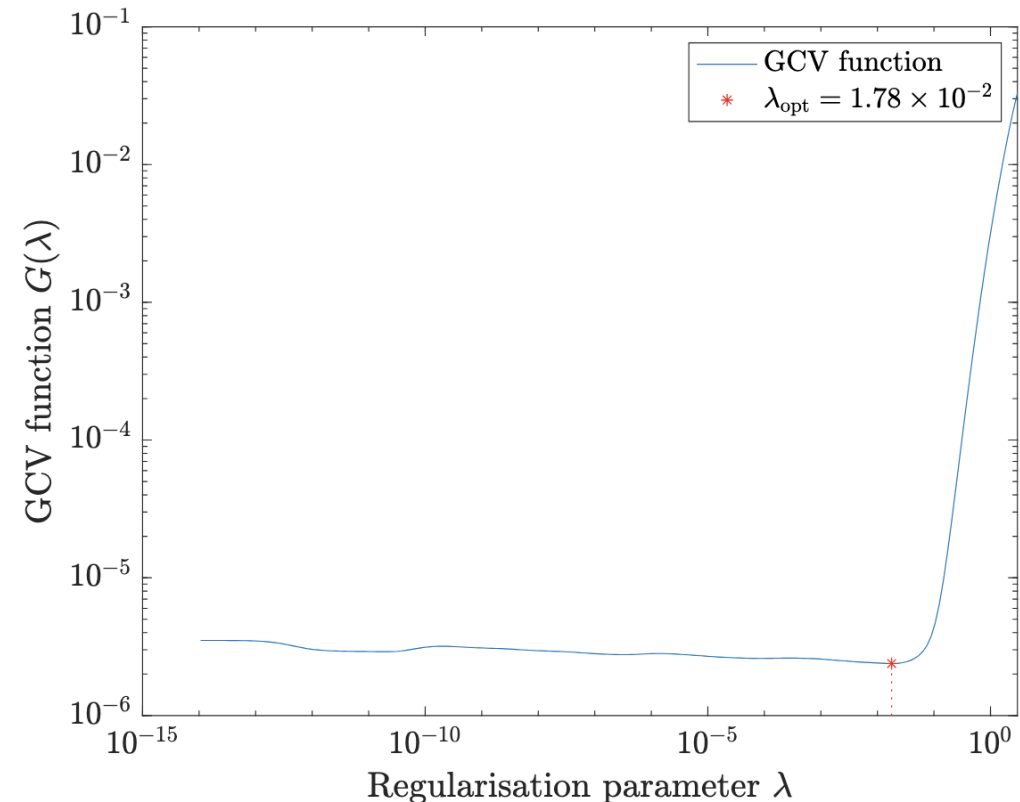
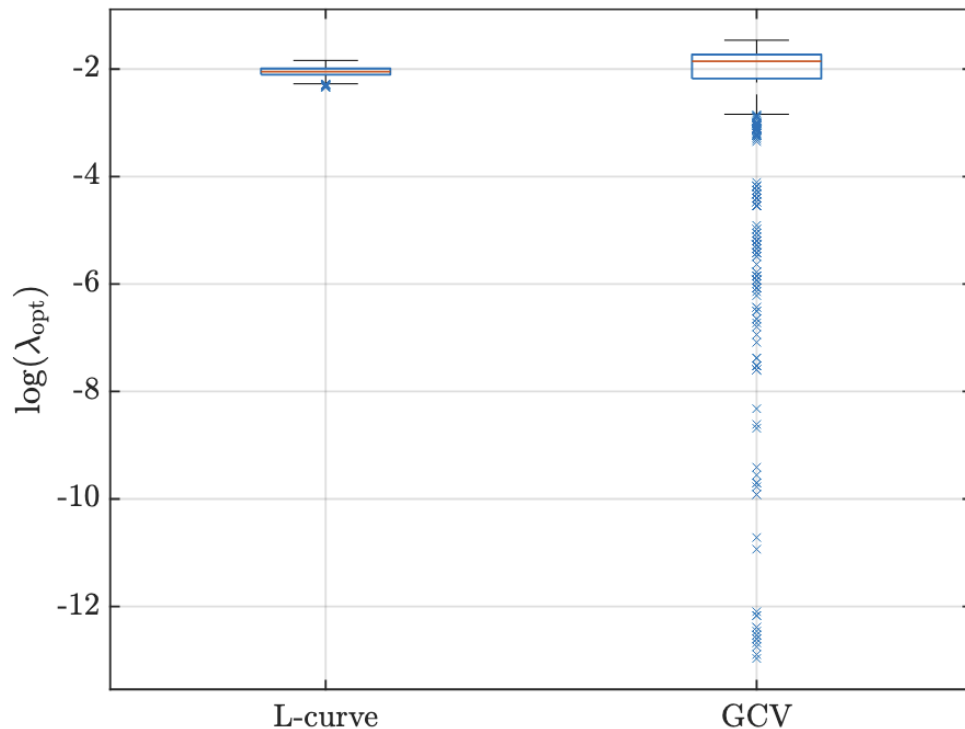
- Water absorption band validates model
- Learning curve of Tecator dataset fails to converge – suggests that it would benefit from more training data



Comparison of Parameter Selection Methods

- L-curve significantly more robust and less conservative than GCV
- L-curve can be used as diagnostic tool unlike GCV

Method	Mean λ_{opt}	Standard deviation	Coefficient of variation	Outliers
L-curve	9.04×10^{-3}	1.74×10^{-3}	19.2%	13
GCV	1.31×10^{-2}	8.22×10^{-3}	62.7%	170



Conclusion

- Replaced blind application with rigorous diagnosis step using the mathematical criteria for regularisation, an L-curve with a distinct corner
- Successfully treated the Tecator dataset which greatly improved accuracy and stability, with the lasso performing best
- Showed the L-curve to be significantly more robust and less conservative parameter selection method than the GCV, and explained why

Future Work

- Explore automating diagnosis step to categorise problem with the user inputting **A** and **b** and the output is **Yes** or **No** – using foundations built by Hansen
- Automation could extend to selecting regularisation technique and parameter based on SVD profile, system dimensions, multicollinearity (VIF)

Thank you, any questions?

References

- **Hansen, P.C.** (2007) 'Regularization tools version 4.1 for Matlab 7.3', *Numerical Algorithms*, 46(2), pp. 189–194.
- **Pace, R.K. and Barry, Ronald** (1997) 'Sparse spatial autoregressions', *Statistics & Probability Letters*, 33(3), pp. 291–297.
- **StatLib Datasets Archive** (1995) *Tecator Meat Sample Dataset*. Available at: <https://lib.stat.cmu.edu/datasets/tecator>.